

Your name (first and last):

Names of people you are working with today (first only is OK):

Part I: Hypotheses

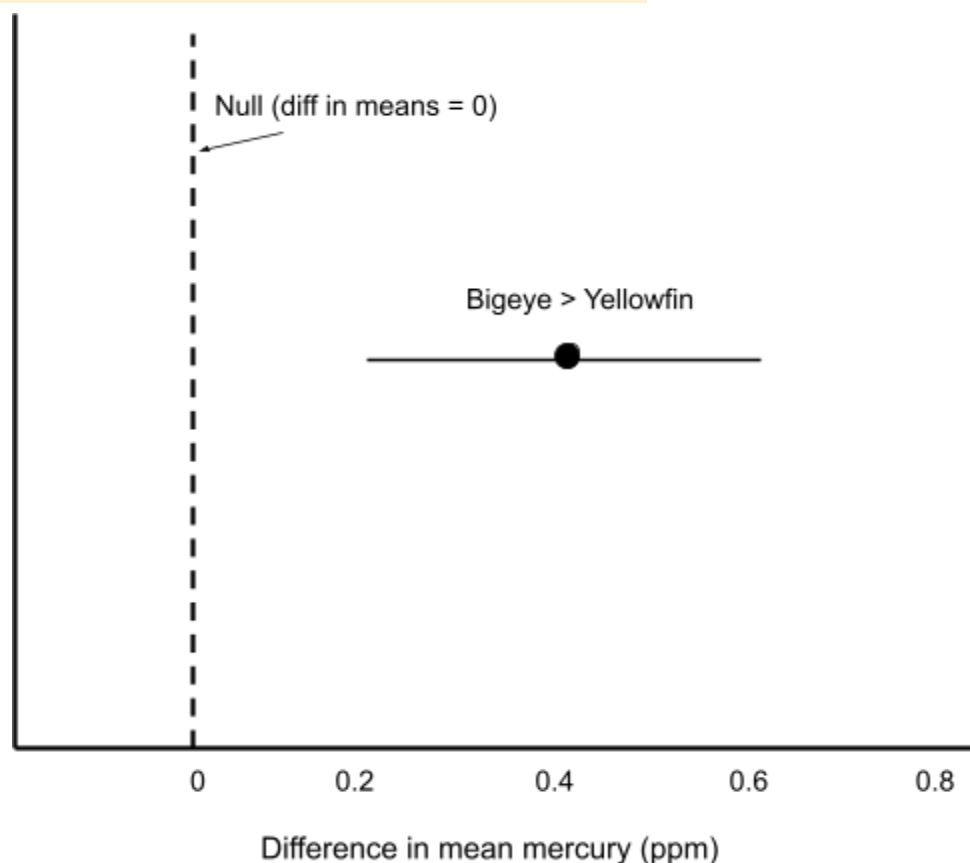
Write out the biological and null hypotheses associated with the research question.

Biological hypothesis: there is an effect of species on mercury (there is a difference in mean mercury between species)

Null hypothesis: there is no effect of species on mercury (any observed difference is due to sampling error... chance)

Part II: Effect plot - visualize the data (info will be provided in lecture slides)

Add numbers along the x-axis; label components as needed so a viewer can understand the figure. Students will need to calculate the difference in means, SE and CI ... math is in lecture slides.



Part III: calculate a p-value

What is the probability of the observed difference in means being due to sampling error (= random chance)? Use the space below to show your work.

8 differences in mean under the null hypothesis were equal to or greater than the actual difference in means (see lecture slides) = $8/100000 = 0.00008$

The probability of the observed difference in means being due to sampling error is 0.008%.

Part IV: Make a claim

Make a claim about mercury in different species of tuna (AKA, the effect of species on mercury). Do yellowfin and bigeye tuna, on average, have different amounts of mercury? Use evidence - specific elements of the figure as well as the p-value - to support your claim.

Bigeye tuna contain more mercury than yellowfin (effect = ~ 0.4 ppm more mercury); the 95% confidence interval does not include zero so we can reject the null hypothesis. There is a very low probability of the observed result being due to chance ($p = 0.00008$). Based on all of this information: there is a strong effect of species on mercury, and the difference in means is statistically significant.